MILITARY SPECIFICATION SHEET

-ADAPTERS, COAXIAL TO WAVEGUIDE (DOUBLE RIDGE) (BANDWIDTH RATIO 2.4:1) (SERIES SMA TO WAVEGUIDE - CLASS 5)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the adapters described herein shall consist of this document and the latest issue of Specification MIL-A-22641.

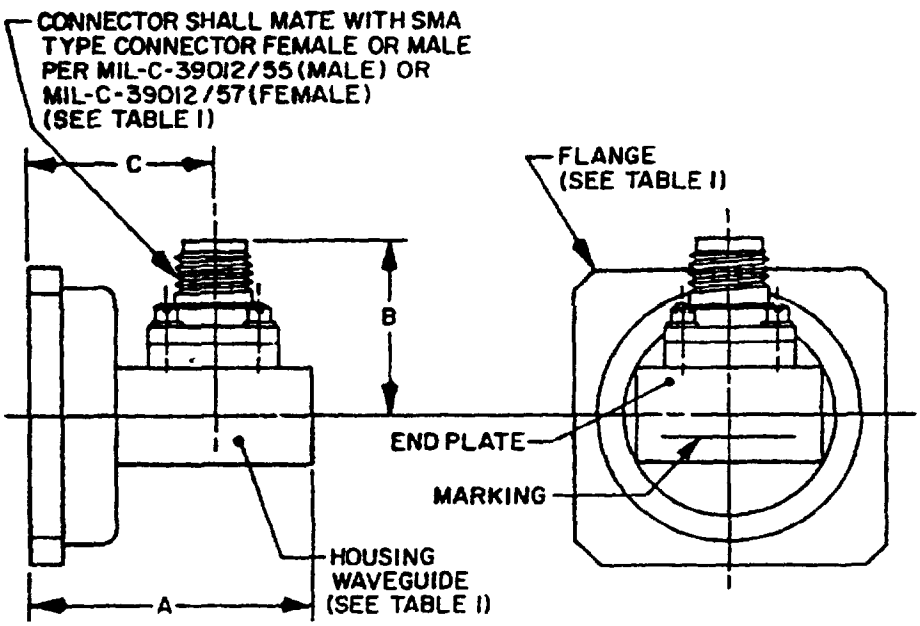


FIGURE 1. Adapters, coaxial to double ridge waveguide (class 5).

REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1 and table I.

Operating temperature range: -54° to +125°C.

Frequency range. See table I.

VSWR: See table I

FSC 5985

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TABLE I. Adapters, coaxial to waveguide (class 5 - SMA to waveguide).

Dash number	Frequency range (GHz)	Typical VSWR over any octave band within frequency range	VSWR over entire frequency range (max) 4/5/	RF insertion loss dB (max) 5/	Waveguide housing	Flange	Waveguide and flange material	Connector	Dimensions 1/2/3/		
									A (max)	B (max)	C (max)
01	.84-2 0	1.10·1	1.15·1	.10	M23351/ 4-017	M39000/ 3-001	Aluminum	Male	4.00 (10.2)	2.20 (56.0)	2.90 (74.0)
02	.84-2.0	1.10·1	1 15:1	10	M23351/ 4-017	M39000/ 3-001	Aluminum	Female	4.00 (10.2)	2.20 (56.0)	2.90 (74.0)
03	2.0-4.8	1.06·1	1.20:1	10	M23351/ 4-025	M39000/ 3-024	Aluminum	Male	2.25 (57.0)	1.20 (30.0)	1.57 (40.0)
04	2.0-4.8	1.06 1	1.20 1	10	M23351/ 4-025	M39000/ 3-024	Aluminum	Female	2.25 (57.0)	1.20 (30.0)	1.57 (40.0)
05	2 0-4.8	1.06·1	1 20.1	10	M23351/ 4-025	M39000/ 3-026	Aluminum	Male	2.25 (57.0)	1.20 (30.0)	1.57 (40.0)
06	2.0-4 8	1 06·1	1 20:1	10	M23351/ 4-025	M39000/ 3-026	Aluminum	Female	2.25 (57.0)	1.20 (30.0)	1.57 (40.0)
07	2.0-4.8	1 06·1	1 20 1	.10	M23351/ 4-025	M39000/ 3-028	Aluminum	Male	2.25 (57.0)	1.20 (30.0)	1.57 (40.0)
08	2 0-4 8	1.06·1	1 20.1	.10	M23351/ 4-025	M39000/ 3-028	Aluminum	Female	2.25 (57.0)	1 20 (30.0)	1.57 (40.0)
09	3.5-8.2	1.06·1	1.15:1	10	M23351/ 4-029	M39000/ 3-030	Aluminum	Male	1.45 (37.0)	.97 (25.0)	.92 (23.0)
10	3.5-8.2	1.06:1	1.15·1	.10	M23351/ 4-029	M39000/ 3-030	Aluminum	Female	1.45 (37.0)	.97 (25.0)	.92 (23.0)
11	3.5-8.2	1.06:1	1.15·1	.10	M23351/ 4-029	M39000/ 3-032	Aluminum	Male	1.45 (37.0)	.97 (25.0)	.92 (23.0)
12	3.5-8.2	1 06·1	1.15:1	10	M23351/ 4-029	M39000/ 3-032	Aluminum	Female	1.45 (37.0)	.97 (25.0)	.92 (23.0)
13	3.5-8.2	1.06 1	1.15.1	.10	M23351/ 4-029	M39000/ 3-034	Aluminum	Male	1.45 (37.0)	.97 (25.0)	.92 (23.0)
14	3.5-8.2	1.06 1	1.15·1	.10	M23351/ 4-029	M39000/ 3-034	Aluminum	Female	1.45 (37.0)	.97 (25.0)	.92 (23.0)
15	4 75-11.0	1.06·1	1.15 1	.10	M23351/ 4-033	M39000/ 3-036	Aluminum	Male	1.75 (44.0)	.90 (23 0)	1.20 (30.0)
16	4 75-11.0	1 06 1	1.15.1	.10	M23351/ 4-033	M39000/ 3-036	Aluminum	Female	1.75 (44.0)	.90 (23.0)	1.20 (30.0)
17	4 75-11.0	1 06:1	1.15 1	.10	M23351/ 4-033	M39000/ 3-038	Aluminum	Male	1.75 (44.0)	.90 (23.0)	1.20 (30.0)

See footnotes at end of table.

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TABLE I. Adapters, coaxial to waveguide (class 5 - SMA to waveguide) - Continued.

Dash number	Frequency range (GHz)	Typical VSWR over any octave band within frequency range	VSWR over entire frequency range (max) 4/5/	RF insertion loss dB (max) 5/	Waveguide housing	Flange	Waveguide and flange material	Connector	Dimensions 1/2/3/		
									A (max)	B (max)	C (max)
18	4.75-11.0	1.06:1	1.15:1	10	M23351/4-033	M39000/3-038	Aluminum	Female	1.75 (44.0)	.90 (23.0)	1.20 (30.0)
19	4.75-11.0	1.06:1	1.15:1	.10	M23351/4-033	M39000/3-040	Aluminum	Male	1.75 (44.0)	.90 (23.0)	1.20 (30.0)
20	4.75-11.0	1.06:1	1.15:1	.10	M23351/4-033	M39000/3-040	Aluminum	Female	1.75 (44.0)	.90 (23.0)	1.20 (30.0)
21	7.5-18.0	1.10:1	1.20:1	.10	M23351/4-037	M39000/3-072	Aluminum	Male	1.30 (33.0)	.75 (19.0)	.77 (20.0)
22	7.5-18.0	1.10:1	1.20:1	.10	M23351/4-037	M39000/3-072	Aluminum	Female	1.30 (33.0)	.75 (19.0)	.77 (20.0)
23	7.5-18.0	1.10:1	1.20:1	10	M23351/4-037	M39000/3-074	Aluminum	Male	1.30 (33.0)	.75 (19.0)	.77 (20.0)
24	7.5-18.0	1.10:1	1.20:1	10	M23351/4-037	M39000/3-074	Aluminum	Female	1.30 (33.0)	.75 (19.0)	.77 (20.0)
25	7.5-18.0	1.10:1	1.20:1	.10	M23351/4-037	M39000/3-076	Aluminum	Male	1.30 (33.0)	.75 (19.0)	.77 (20.0)
26	7.5-18.0	1.10:1	1.20:1	10	M23351/4-037	M39000/3-076	Aluminum	Female	1.30 (33.0)	.75 (19.0)	.77 (20.0)

1/ Dimensions are in inches.
2/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.
4/ A larger attenuator pad may be used for VSWR test setup (see figure 1 of MIL-A-22641)
5/ VSWR and RF insertion loss may be measured by swept frequency techniques

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REQUIREMENTS - Continued

RF insertion loss See table I

Connector Connector bodies shall be corrosion-resistant steel.

Flange, waveguide See table I

Housing, waveguide. See table I.

Marking. See figure 1

Part number M22641/22- (dash number from table I).

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities.
Army - MI, EL, GL
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR
Navy - AS, MC

Preparing activity:
Navy - EC

Agent
DLA - ES

(Project 5985-0876-3)

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